Introduction to

# Artificial Intelligence

with Python

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# Knowledge

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# knowledge-based agents

agents that reason by operating on internal representations of knowledge

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**If it didn't rain, Harry visited Hagrid today.**

Harry visited Hagrid or Dumbledore today, but not both.

Harry visited Dumbledore today.

**Harry did not visit Hagrid today.**

**It rained today.**

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# Logic

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# sentence

an assertion about the world
in a knowledge representation language

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# Propositional Logic

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# Proposition Symbols

$P$ $Q$ $R$

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# Logical Connectives

| ¬ | ∧ | ∨ |
| --- | --- | --- |
| not | and | or |

| → | ↔ |
| --- | --- |
| implication | biconditional |

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# Not ($\neg$)

| $P$ | $\neg P$ |
| --- | --- |
| false | true |
| true | false |

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# And ($\wedge$)

| $P$ | $Q$ | $P \wedge Q$ |
|---|---|---|
| false | false | false |
| false | true | false |
| true | false | false |
| true | true | true |

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# Or (v)

| $P$ | $Q$ | $P \vee Q$ |
|---|---|---|
| false | false | false |
| false | true | true |
| true | false | true |
| true | true | true |

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# Implication ($\rightarrow$)

| $P$ | $Q$ | $P \rightarrow Q$ |
|---|---|---|
| false | false | true |
| false | true | true |
| true | false | false |
| true | true | true |

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# Biconditional ($\leftrightarrow$)

| $P$ | $Q$ | $P \leftrightarrow Q$ |
|---|---|---|
| false | false | true |
| false | true | false |
| true | false | false |
| true | true | true |

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# model

assignment of a truth value to every propositional symbol (a "possible world")

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## model

$P$: It is raining.

$Q$: It is a Tuesday.

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$$\{P = \texttt{true}, Q = \texttt{false}\}$$

# knowledge base

a set of sentences known by a knowledge-based agent

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# Entailment

$$\alpha \models \beta$$

In every model in which sentence **α** is true, sentence $\beta$ is also true.

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**If it didn't rain, Harry visited Hagrid today.**

Harry visited Hagrid or Dumbledore today, but not both.

Harry visited Dumbledore today.

**Harry did not visit Hagrid today.**

**It rained today.**

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# inference

the process of deriving new sentences from old ones

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$P$: It is a Tuesday.

$Q$: It is raining.

$R$: Harry will go for a run.

KB: $(P \wedge \neg Q) \rightarrow R$ $P$ $\neg Q$

Inference: $R$

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# Inference Algorithms

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Does

$$\mathrm{KB} \models \alpha$$

?

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# Model Checking

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# Model Checking

- To determine if $\mathbf{KB} \models \alpha$:
  - Enumerate all possible models.
  - If in every model where $\mathbf{KB}$ is true, $\alpha$ is true, then $\mathbf{KB}$ entails $\alpha$.
  - Otherwise, $\mathbf{KB}$ does not entail $\alpha$.

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$P$: It is a Tuesday. $Q$: It is raining. $R$: Harry will go for a run.

KB: $(P \wedge \neg Q) \rightarrow R$ $P$ $\neg Q$

Query: $R$

| $P$ | $Q$ | $R$ | KB |
|---|---|---|---|
| false | false | false | |
| false | false | true | |
| false | true | false | |
| false | true | true | |
| true | false | false | |
| true | false | true | |
| true | true | false | |
| true | true | true | |

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$P$: It is a Tuesday. $Q$: It is raining. $R$: Harry will go for a run.

KB: $(P \wedge \neg Q) \rightarrow R$ $P$ $\neg Q$

Query: $R$

| $P$ | $Q$ | $R$ | KB |
|---|---|---|---|
| false | false | false | false |
| false | false | true | false |
| false | true | false | false |
| false | true | true | false |
| true | false | false | false |
| true | false | true | true |
| true | true | false | false |
| true | true | true | false |

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$P$: It is a Tuesday. $Q$: It is raining. $R$: Harry will go for a run.

KB: $(P \wedge \neg Q) \rightarrow R$ $P$ $\neg Q$

Query: $R$

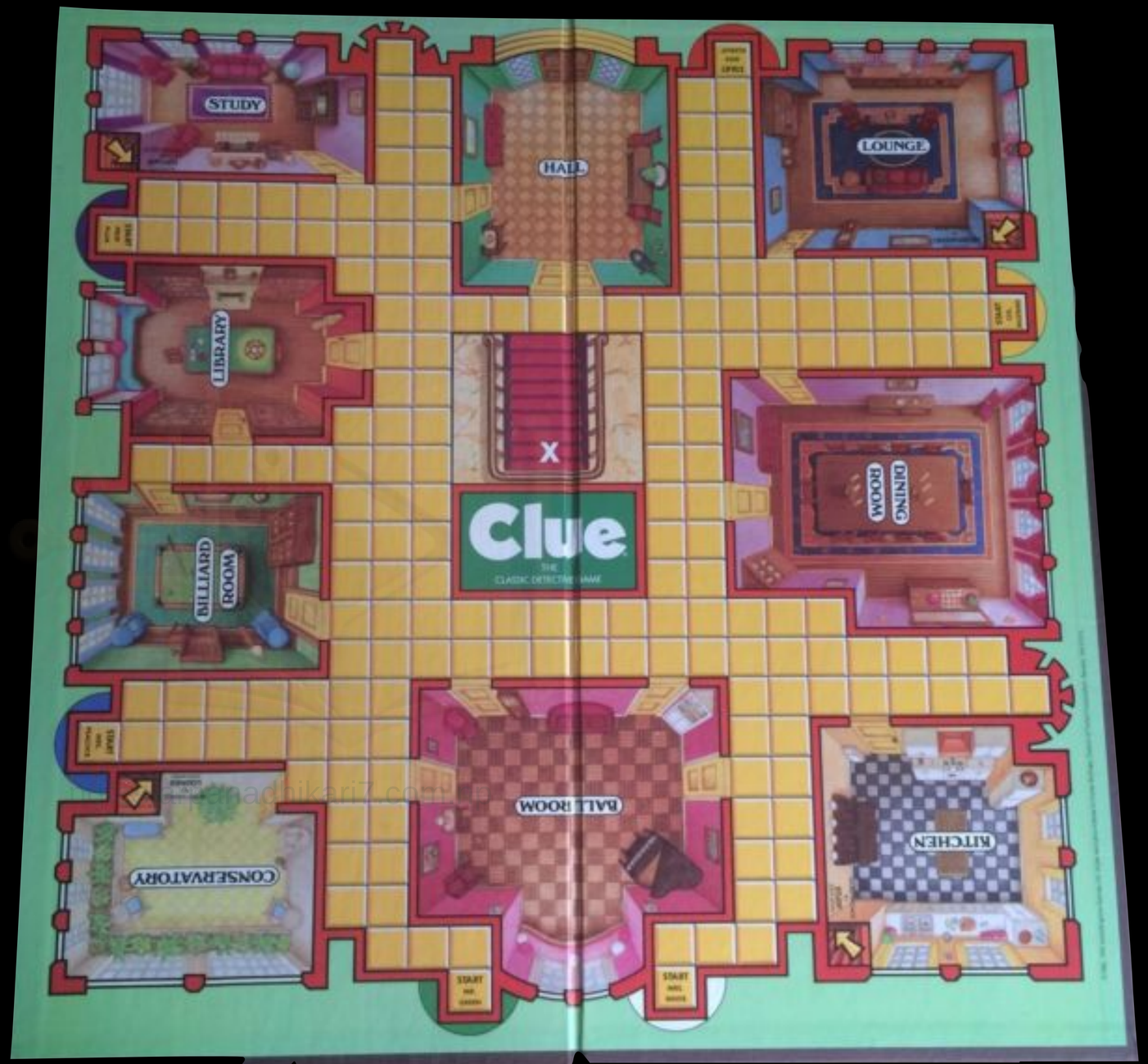
| $P$ | $Q$ | $R$ | KB |
|---|---|---|---|
| false | false | false | false |
| false | false | true | false |
| false | true | false | false |
| false | true | true | false |
| true | false | false | false |
| true | false | true | true |
| true | true | false | false |
| true | true | true | false |

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# Knowledge Engineering

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# Clue




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# Clue

| People | Rooms | Weapons |
| --- | --- | --- |
| Col. Mustard | Ballroom | Knife |
| Prof. Plum | Kitchen | Revolver |
| Ms. Scarlet | Library | Wrench |

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# Clue

| People | Rooms | Weapons |
| --- | --- | --- |
| | | |
| | | |

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# Clue

| People | Rooms | Weapons |
| --- | --- | --- |
| | | |
| | | |

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People

Rooms

Weapons

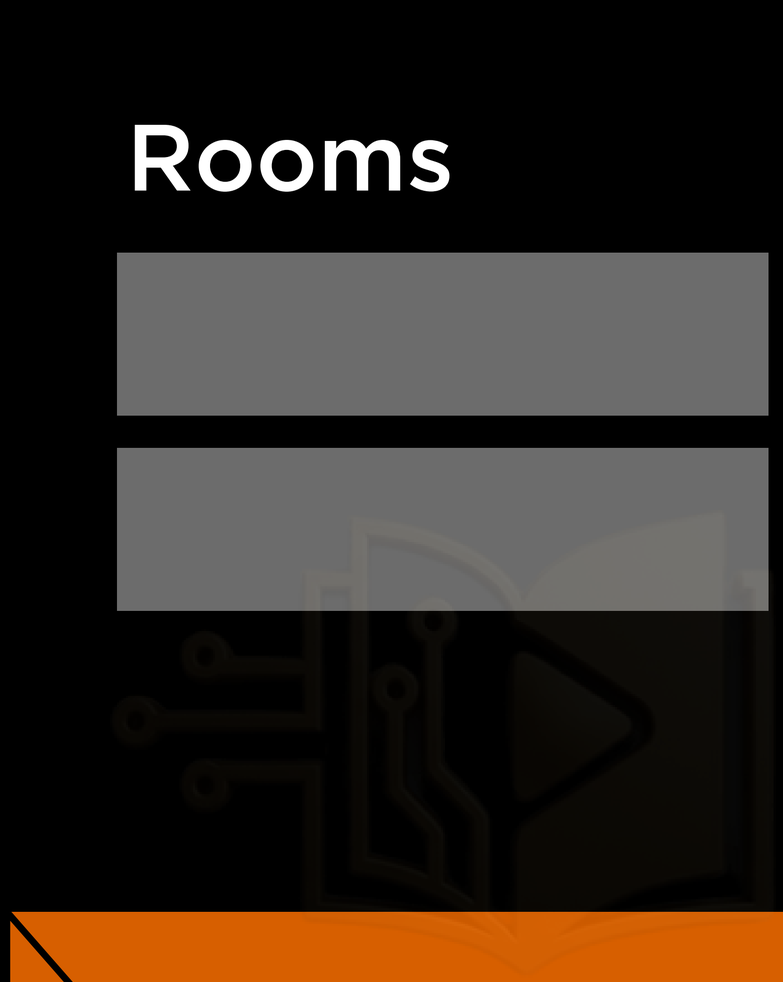
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People

Rooms

Weapons



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# Clue

## Propositional Symbols

| | | |
|---|---|---|
| *mustard* | *ballroom* | *knife* |
| *plum* | *kitchen* | *revolver* |
| *scarlet* | *library* | *wrench* |

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# Clue

$$(mustard \vee plum \vee scarlet)$$

$$(ballroom \vee kitchen \vee library)$$

$$(knife \vee revolver \vee wrench)$$

$$\neg plum$$

$$\neg mustard \vee \neg library \vee \neg revolver$$

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# Logic Puzzles

- Gilderoy, Minerva, Pomona and Horace each belong to a different one of the four houses: Gryffindor, Hufflepuff, Ravenclaw, and Slytherin House.
- Gilderoy belongs to Gryffindor or Ravenclaw.
- Pomona does not belong in Slytherin.
- Minerva belongs to Gryffindor.

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# Logic Puzzles

## Propositional Symbols

*GilderoyGryffindor*
*GilderoyHufflepuff*
*GilderoyRavenclaw*
*GilderoySlytherin*

*MinervaGryffindor*
*MinervaHufflepuff*
*MinervaRavenclaw*
*MinervaSlytherin*

*PomonaGryffindor*
*PomonaHufflepuff*
*PomonaRavenclaw*
*PomonaSlytherin*

*HoraceGryffindor*
*HoraceHufflepuff*
*HoraceRavenclaw*
*HoraceSlytherin*

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# Logic Puzzles

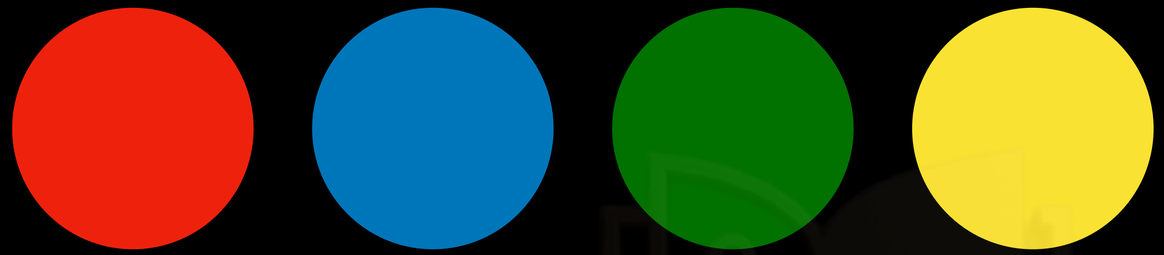
$$(PomonaSlytherin \rightarrow \neg PomonaHufflepuff)$$

$$(MinervaRavenclaw \rightarrow \neg GilderoyRavenclaw)$$

$$(GilderoyGryffindor \vee GilderoyRavenclaw)$$

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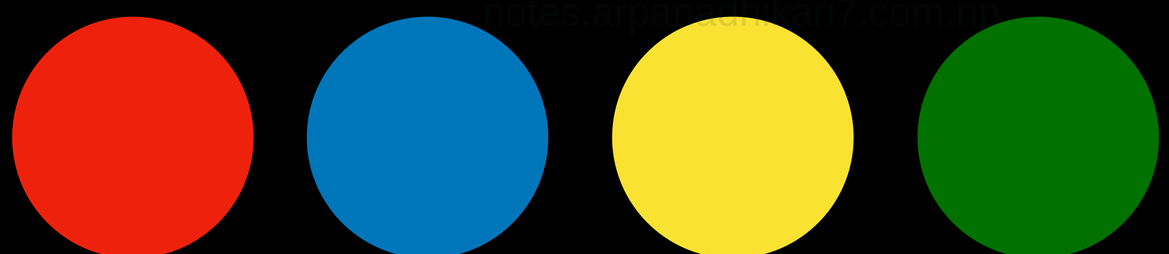
# Mastermind



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# Inference Rules

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# Modus Ponens

If it is raining, then Harry is inside.

It is raining.

---

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Harry is inside.

# Modus Ponens

$$\frac{\alpha \rightarrow \beta \qquad \alpha}{\beta}$$

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# And Elimination

Harry is friends with Ron and Hermione.

---

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Harry is friends with Hermione.

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# And Elimination

$$\frac{\alpha \wedge \beta}{\alpha}$$

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# Double Negation Elimination

It is not true that Harry did not pass the test.

---

Harry passed the test.

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# Double Negation Elimination

$$\frac{\neg(\neg\alpha)}{\alpha}$$

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# Implication Elimination

If it is raining, then Harry is inside.

---

It is not raining or Harry is inside.

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# Implication Elimination

$$\frac{\alpha \rightarrow \beta}{\neg \alpha \vee \beta}$$

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# Biconditional Elimination

It is raining if and only if Harry is inside.

---

If it is raining, then Harry is inside,
and if Harry is inside, then it is raining.

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# Biconditional Elimination

$$\alpha \leftrightarrow \beta$$

---

$$(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$$

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# De Morgan's Law

It is not true that both
Harry and Ron passed the test.

---

Harry did not pass the test
or Ron did not pass the test.

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# De Morgan's Law

$$\neg(\alpha \wedge \beta)$$

---

$$\neg\alpha \vee \neg\beta$$

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# De Morgan's Law

It is not true that
Harry or Ron passed the test.

---

Harry did not pass the test
and Ron did not pass the test.

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# De Morgan's Law

$$\neg(\alpha \vee \beta)$$

---

$$\neg\alpha \wedge \neg\beta$$

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# Distributive Property

$$(\alpha \wedge (\beta \vee \gamma))$$

---

$$(\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$$

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# Distributive Property

$$(\alpha \vee (\beta \wedge \gamma))$$

---

$$(\alpha \vee \beta) \wedge (\alpha \vee \gamma)$$

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# Search Problems

- initial state
- actions
- transition model
- goal test
- path cost function

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# Theorem Proving

- initial state: starting knowledge base
- actions: inference rules
- transition model: new knowledge base after inference
- goal test: check statement we're trying to prove
- path cost function: number of steps in proof

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# Resolution

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(Ron is in the Great Hall) v (Hermione is in the library)

Ron is not in the Great Hall

---

Hermione is in the library

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$$P \vee Q$$

$$\neg P$$

---

$$Q$$

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$$P \vee Q_1 \vee Q_2 \vee \ldots \vee Q_n$$

$$\neg P$$

---

$$Q_1 \vee Q_2 \vee \ldots \vee Q_n$$

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(Ron is in the Great Hall) v (Hermione is in the library)

(Ron is not in the Great Hall) v (Harry is sleeping)

---

(Hermione is in the library) v (Harry is sleeping)

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$$P \vee Q$$

$$\neg P \vee R$$

---

$$Q \vee R$$

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$$P \vee Q_1 \vee Q_2 \vee \ldots \vee Q_n$$

$$\neg P \vee R_1 \vee R_2 \vee \ldots \vee R_m$$

---

$$Q_1 \vee Q_2 \vee \ldots \vee Q_n \vee R_1 \vee R_2 \vee \ldots \vee R_m$$

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# clause

a disjunction of literals

e.g. $P \vee Q \vee R$

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# conjunctive normal form

logical sentence that is a conjunction of clauses

e.g. $(A \vee B \vee C) \wedge (D \vee \neg E) \wedge (F \vee G)$

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# Conversion to CNF

- Eliminate biconditionals
    - turn $(\alpha \leftrightarrow \beta)$ into $(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$
- Eliminate implications
    - turn $(\alpha \rightarrow \beta)$ into $\neg\alpha \vee \beta$
- Move $\neg$ inwards using De Morgan's Laws
    - e.g. turn $\neg(\alpha \wedge \beta)$ into $\neg\alpha \vee \neg\beta$
- Use distributive law to distribute $\vee$ wherever possible

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# Conversion to CNF

$$(P \vee Q) \rightarrow R$$

$$\neg(P \vee Q) \vee R$$ eliminate implication

$$(\neg P \wedge \neg Q) \vee R$$ De Morgan's Law

$$(\neg P \vee R) \wedge (\neg Q \vee R)$$ distributive law

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# Inference by Resolution

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$$P \vee Q$$

$$\neg P \vee R$$

---

$$(Q \vee R)$$

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$$P \vee Q \vee S$$

$$\neg P \vee R \vee S$$

---

$$(Q \vee S \vee R \vee S)$$

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$$P \vee Q \vee S$$

$$\neg P \vee R \vee S$$

---

$$(Q \vee R \vee S)$$

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$$P$$

$$\neg P$$

---

$$( )$$

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# Inference by Resolution

- To determine if $\mathbf{KB} \models \alpha$:
  - Check if $(\mathbf{KB} \land \neg\alpha)$ is a contradiction?
    - If so, then $\mathbf{KB} \models \alpha$.
    - Otherwise, no entailment.

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# Inference by Resolution

- To determine if $\mathbf{KB} \models \alpha$:
  - Convert $(\mathbf{KB} \wedge \neg\alpha)$ to Conjunctive Normal Form.
  - Keep checking to see if we can use resolution to produce a new clause.
    - If ever we produce the **empty** clause (equivalent to False), we have a contradiction, and $\mathbf{KB} \models \alpha$.
    - Otherwise, if we can't add new clauses, no entailment.

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A)$$

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad \underline{(\neg B \vee C)} \quad \underline{(\neg C)} \quad (\neg A)$$

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad \underline{(\neg B \vee C)} \quad \underline{(\neg C)} \quad (\neg A) \quad (\neg B)$$

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A) \quad (\neg B)$$

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$\underline{(A \vee B)} \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A) \quad \underline{(\neg B)}$$

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$\underline{(A \vee B)} \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A) \quad \underline{(\neg B)} \quad (A)$$

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$(A \vee B)$ $(\neg B \vee C)$ $(\neg C)$ $(\neg A)$ $(\neg B)$ $(A)$

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad \underline{(\neg A)} \quad (\neg B) \quad \underline{(A)}$$

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad \underline{(\neg A)} \quad (\neg B) \quad \underline{(A)} \quad ( )$$

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# Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail $A$?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A) \quad (\neg B) \quad (A) \quad ( )$$

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# First-Order Logic

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# Propositional Logic

## Propositional Symbols

*MinervaGryffindor*

*MinervaHufflepuff*

*MinervaRavenclaw*

*MinervaSlytherin*

...

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# First-Order Logic

Constant Symbol

*Minerva*
*Pomona*
*Horace*
*Gilderoy*
*Gryffindor*
*Hufflepuff*
*Ravenclaw*
*Slytherin*

Predicate Symbol

*Person*
*House*
*BelongsTo*

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# First-Order Logic

$Person(Minerva)$ Minerva is a person.

$House(Gryffindor)$ Gryffindor is a house.

$\neg House(Minerva)$ Minerva is not a house.

$BelongsTo(Minerva, Gryffindor)$

Minerva belongs to Gryffindor.

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# Universal Quantification

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# Universal Quantification

$$\forall x.\ BelongsTo(x,\ Gryffindor) \rightarrow \neg BelongsTo(x,\ Hufflepuff)$$

For all objects x, if x belongs to Gryffindor,
then x does not belong to Hufflepuff.

Anyone in Gryffindor is not in Hufflepuff.

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# Existential Quantification

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# Existential Quantification

$\exists x.\ House(x) \wedge BelongsTo(Minerva,\ x)$

There exists an object x such that
x is a house and Minerva belongs to x.

Minerva belongs to a house.

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# Existential Quantification

$$\forall x.\ Person(x) \rightarrow\ (\exists y.\ House(y) \wedge BelongsTo(x, y))$$

For all objects x, if x is a person, then
there exists an object y such that
y is a house and x belongs to y.

Every person belongs to a house.

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